

Gedächtnisprotokoll

Data Literacy

bei Prof. Hennig

9. März 2021

Klausur:	Hauptklausur, WS2020/21
Pruefer:	Prof. Hennig
Datum:	02.03.2021
Zeit:	90 min
Punkte:	100
Hilfsmittel:	keine
Sprache:	Englisch
Modul:	ML4102

1 Question 1 - 6 Punkte

Description of a study: several thousand randomly chosen people with attributes x_i describing features like gender, age, etc. The people with attributes $X = \{x_1, \dots, x_N\}$ were divided into two groups $g_1, g_2 \subset [1, \dots, N]$. The people in one group received a vaccine ($z_i = 1$), the people in the other group received a placebo ($z_i = 0$), and it was monitored whether they contracted the disease (d_i).

What kind of study is this? Observational, active, metastudy? (3 Punkte)

Which one is the independent variable, which one is the dependent variable? (3 Punkte)

2 Question 2 - 18 Punkte (?)

Write down the definition of the information content of the event $X = x_i$ and $Y = y_i$. (5 Punkte)

Write down the definition of the entropy of the events X and Y . (5 Punkte)

Proof that for two independent random variables X and Y , the entropy of the joint distribution is $\mathbb{H}(p(X, Y)) = \mathbb{H}(p(X)) + \mathbb{H}(p(Y))$. (8 Punkte)

3 Question 3 - 12 Punkte (?)

In a country with a pandemic, there is a rule that stricter measures are to be taken when a county has an incidence above 35 cases per 100000 inhabitants. In a small county with population 63197, there are 32 cases. That means that the incidence rate is 50.6. The mayor of this county complains that there is a disadvantage for his county as it is so small, and that the true incidence rate in his county could actually be 35. Plot: binomial distribution and survival function for $f = 35/100000$ and $n = 63197$ (Figure 1). What is the p-value for the mayor's hypothesis? (It was 0.02) (6 Punkte)

There are 401 counties in the country. For how many would you expect a p-value as small as this one or lower by chance? (6 Punkte)

4 Question 4 - 20 Punkte

For a centered normal distribution: maximum likelihood estimation for σ^2 . (4 Punkte)

Determine the Fisher information $I_n(\sigma^2)$ for this distribution. (12 Punkte)

Use this Fisher information to construct a distribution for the maximum likelihood estimator. (4 Punkte)

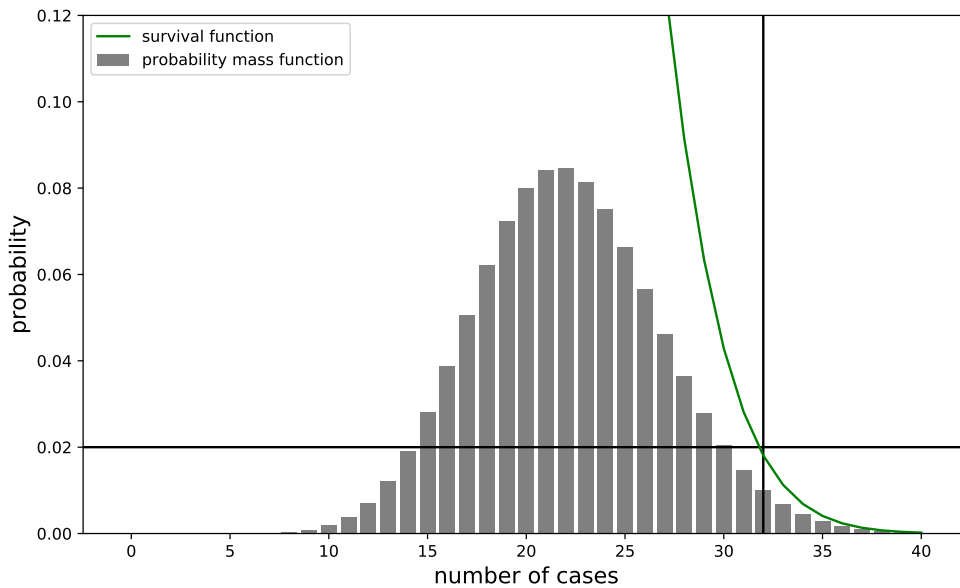


Figure 1: Binomial distribution of cases under the mayor's (null) hypothesis.

5 Question 5 - 12 Punkte

For a data set ($n \times 4$) about motors consisting of four columns with different units, PCA was performed to reduce it to two dimensions. Describe and explain qualitatively what happens with the principal components and associated principal values in the following two scenarios.

Instead of in Hz (rotations per second), the second column was measured in rotations per day. (7 Punkte)

All motors that were measured had 48 slots (the third column was the same for all n). (5 Punkte)

6 Question 6 - 10 Punkte

Name the four high-level stages of the machine learning deployment workflow and briefly describe a key challenge for each of them (one sentence).

7 Question 7 - 10 Punkte

For a given joint distribution $p(A, R, Y)$, does it fulfill the fairness criterion of independence?

8 Question 8 - 12 Punkte

Three plots shown (Figure 2), with different (log) scales, had to guess the function. All of them were straight lines, and you had to look closely at the scales (obviously, the functions were not given).

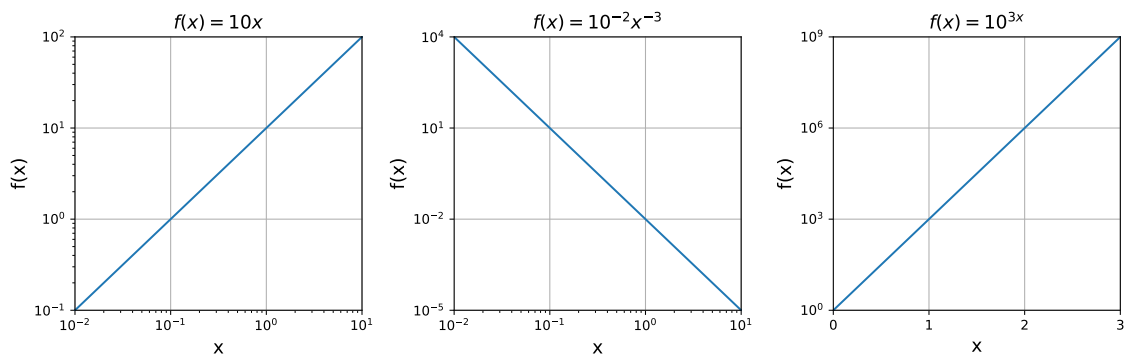


Figure 2: Three different functions.